Datablast Data Engineer Interview – Technical Test

These questions are designed to test your logical thinking, data modelling, stats, and programming ability.

Some pointers:

- You won't need a calculator.
- The whole test can be completed in two hours, but often takes longer. **You will have 5 hours in total**.
- The test should be self-explanatory. You may find some questions easier than others, some are well defined, others much less well defined. Note that this has been done by design.
- Use as much paper as you like – we are interested in how you worked out your answers as well as the answers themselves, so please do include some of your workings.
- An important skill for a Data Scientist is to validate your work, so please check your answers (and where appropriate show how you may have checked them).
- It is absolutely essential that you provide the self-ratings for Question 5.

Once you have completed your test, please email the answers (you can type out your answers on a word document or use pen/paper and email photos of your answers). Please make sure your name is on the front of your answer sheets.

Remember you have **5 hours** to complete your test.

Question 1
Write a simple Python script that:

- fetches the USD currency rates for every available currency between the dates 2021-01-01 and 2021-01-10,
- produces a CSV file with the results.

The result file should look as follows, with a new row for every date and currency pair:

```
date,name,value
2021-01-01,"JPY",113.66631
2021-01-01,"PGK",3.51129
2021-01-01,"TJS",11.29024
2021-01-01,"XPF",105.04141
2021-01-01,"MGA",3976.07617
2021-01-01,"EUR",1.13740
...
```

The requirements are:

- You can use any free currency API out there, it is your choice.
- The script must be written in Python 3.
- The result must be delivered as a Git repository hosted online.
  - Either GitHub or GitLab is fine.
  - You can make the repository public, just make sure you are not exposing any API credentials you might have.
  - As long as you share the repository with us somehow, you can also keep it private.

Question 2
Find the following questions using SQL from the table below:

Table “*CustomerLogins”*

CustomerID	Name	LoginDate	Platform
123456	Joe	2015-06-01	IOS
987654	Donald	2015-05-23	Android
…	…	…	…

2.1. What’s the total number of rows in this table?
2.2. What’s the total number of rows corresponding to logins on 1st May 2015?
2.3. What’s the number of unique customers?
2.4. What statement would be used to create the above table (without any data in it).
2.5. In SQL, what is the difference between an inner join and an outer join? Please give an example of some circumstances where you would recommend using an outer join?

Question 3
SITO elevators have a system to simulate different elevator control mechanisms in their buildings. Let us assume a building has M elevators and N floors. You are in charge of measuring the performance of different elevator control mechanisms and so will need to design the data model to capture observed data and the measurements you would calculate on that data model. Assume a typical simulation run proceeds over a 24-hour period and you are allowed to observe as much as you like (when/where each elevator is, how many passengers, where the passengers are, when/where they arrive/depart, etc – if in doubt assume you can observe it). From your detailed data model, you will then need (with very simple calculations) to determine various performance measures, for example:

- Average waiting time per passenger
- Average journey time per passenger
- etc

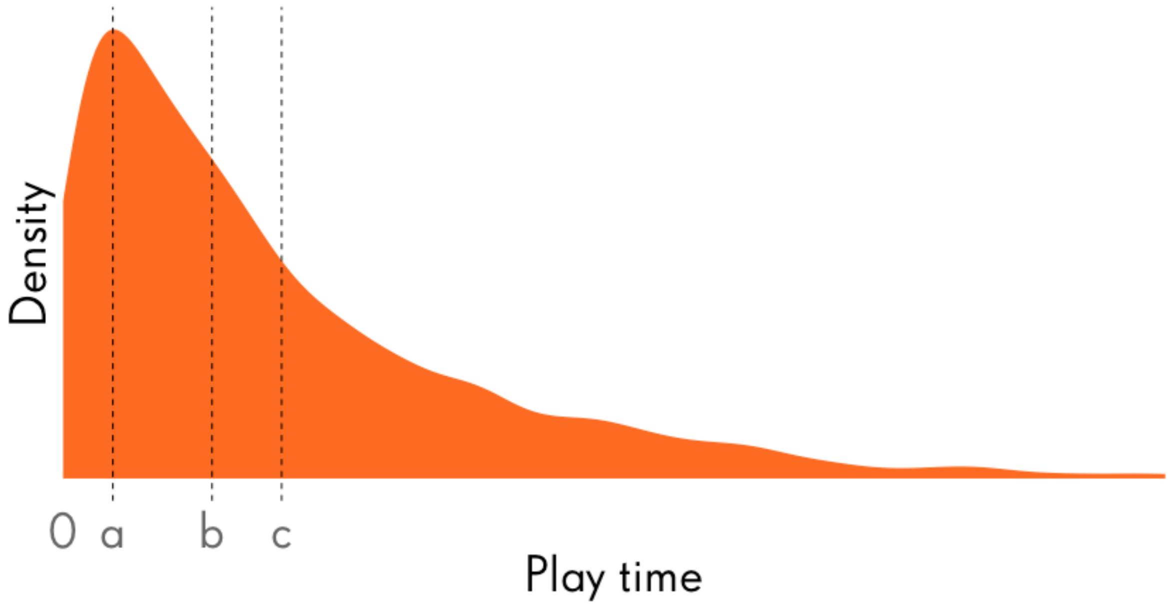
3.1. List the different stakeholders who would be interested in elevator performance (a “stakeholder” is any person or group who have an interest in or may be affected by some aspect of elevator performance).

3.2. List other performance measures that it would be useful or important to measure – make sure these covers all the stakeholders. (Hint: there are lots and lots of these. Aim for 10 or more...).

3.3. What would a suitable data representation look like? Please design a series of tables (as would be suitable to put in a database or spreadsheet). Make sure that the data representation (with very simple arithmetic calculations) is adequate to calculate the above measures, and any other measures that you deem important (and that those calculations are fairly, easy, and unambiguous). Please point out any problems you might expect to arise with your data model.

3.4. For “Average waiting time per passenger” and at least 2 other performance measures, describe how they can be easily calculated from your data model. Preferably write the SQL code you would use to calculate the waiting and journey times.

Question 4
The chart below represents the distribution of play time in a day, for a sample of our players. The points a, b and c mark the values of three measures used to summarize the data.



4.1. What standard measures are a, b and c most likely denoting? Please explain why.

4.2. If you drew n samples from this distribution and measured their mean, then repeated that many times, how would you expect the distribution of those sample means to differ from this distribution?
4.3. Would its standard deviation be bigger, smaller, or the same as this distribution's standard deviation and why?

Question 5
The following two pages contain our “Data Science competency matrix”. Please rate yourself on a scale from 1-5 in each of the areas in the matrix:

1- Coding
2- SQL
3- Stats

Our scale of 1-5 in each area is described in the two pages which follow. Please answer honestly. We certainly don't expect many people to be 4's and 5's in many areas and equally would be surprised if you were not a 1 or 2 in some areas.

	Rating 1	Rating 2	Rating 3	Rating 4	Rating 5
Coding	I can't code. If pressed I might make minor changes to other people's code.	I can write simple programs, but I mostly build on what others have done. I have one or two preferred languages, but am aware there are lots of other things out there I could learn.	I've written thousands of lines of code, using a few standard libraries, in a couple of different languages. I understand basic data structures and algorithms and can use them efficiently. I can probably pick up the basics of new languages fairly quickly, and I actively enjoy learning new tools.	My code is elegantly structured, and I think about the right tools for the right job. I easily work in teams of 5+ people on the same codebase, and I understand the benefits of version control systems. I could code up my own data structures, efficiently, but prefer to use the great libraries others have built. I easily pick up new languages, and can easily work with and extend other people's code. I'm pretty skilled in some of the analytical coding languages/tools used by data scientists (R, python, Matlab, Qlikview, Tableau, Bash,..)	I'm a fantastic software engineer who others look up to. I push others to write better code, share my knowledge of design patterns and keep myself on the cutting edge of new languages, libraries, techniques. I work well with others, and am able to effortlessly refactor complex code2bases to make them more robust and easier to use. I'm very skilled in several of the analytical coding languages/tools used by data scientists (R, python, Matlab, Qlikview, Tableau, Bash,..)
SQL	Excel is the database I use most. Occasionally I've used SQL and can write basic select queries.	I know basic database concepts and am comfortable writing simple SQL selects, and making changes to SQL other people write. Most of the time I can get the data I want using simple group2 by/join/etc constructs. Usually I can't tell if my SQL is going to run efficiently or not, and when things get tricky I need help from a SQL expert.	Almost all the time I can get the data I need with the right joins, selects, group bys, etc 2 usually I can't tell if my SQL is going to run efficiently or not, and I'm actively working to expand my knowledge of more sophisticated SQL constructs, and learning the difference between dialects of SQL (and HQL). I do find it helpful to have a SQL expert around for the tricky situations and to learn from.	I'm a SQL wizard and people always come to me to help them optimise their queries, or just to pick up my old queries to modify for their own purposes. I've picked up most of the subtle differences between HiveQL and SQL and make good use of analytic functions specific to the available language. I know a fair amount about dimensional modelling and how to design good database schemas. I understand the differences between types of database and why to use one over another for a given application.	I'm a data2modelling expert, and regularly share my knowledge of dimensional modelling, star schemas, facts and dimensions. I can build and deploy my own databases, am familiar with the ETL process and can schedule my own jobs as part of that workflow. I work closely with Data Warehouse teams to help them improve the data model so my analytical work is as effortless and accurate as possible. I can effectively work directly with hadoop or other tools to process data when it's more efficient than working through SQL.
Stats	I can use graphs and descriptive statistics to summarise a data set	I understand and know when to use different statistical charts such as histograms and scatter plots. I can build linear models for two or more variables, understand error bars, standard deviations, and use standard software to calculate statistical significance when necessary	I am familiar with the concepts of inferential statistics and know how to perform different parametric tests appropriate to my experimental design. I am very familiar with various statistical distributions and successfully utilise this knowledge in my analysis and modelling. I can run standard machine learning algorithms from libraries using tools like R & Python, typically using what I think is the right methodology for the right problem.	I've got a great understanding of probability theory and Bayesian statistics and can productively debate the pros/cons of different statistical approaches.I am familiar with parametric and non parametric tests and their appropriate usage. I am quite versed in techniques such as boostrapping, modelling of covariates, and training and fitting of machine learning models. I know how to operate with analysis of time series and concepts such as cointegration and autoregression are very familiar to me. I understand how to use power calculations to help design experiments that will produce reliable results.	I am an expert in in classical and Bayesian statistics, fully understanding the differences between the approaches and am able to operate within both as required. I am familiar with Experimental Design and work with others to create experiments that avoid biases and allow for successful statistical inference. I develop my own statistical models for custom analysis and can program statistical procedures to understand unfamiliar situations, using techniques such as bootstrap resampling and Markov Chain Monte Carlo simulations. I regularly read the literature for developments in statistical modelling which I can put into practice.

Question 6
What does your favorite workplace look like?